

B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61512

Course Code: SH/CSC/602/C-14

**Computer Graphics
(CBCS OLD 2017-18)**

Time: 1 Hr 15 Min

Full Marks: 25

*The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.*

Answer the questions unit wise as instructed

UNIT- I

1. Answer any five of the following questions: $5 \times 1 = 5$

- a) What do you mean by vertical retrace of the electron beam?
- b) What is frame buffer?
- c) Write two examples of scan converted objects.
- d) What is anti-aliasing?
- e) What is the role of electron gun in a CRT monitor?
- f) Name a hidden surface removal algorithm.
- g) What are the different types of clipping?
- h) Name a rigid body transformation that rotates objects without deformation.

UNIT - II

2. Answer any two of the following questions: $2 \times 5 = 10$

- a) Write algorithm to fill a polygon using Boundary fill.
- b) Consider the line from (0, 0) to (5, 6). Use DDA algorithm to rasterize this line.
- c) Write an algorithm to generate Bezier curve.

[Turn Over]

- d) Obtain the composite transformation matrix for general fixed point rotation.

UNIT - III

3. Answer any one of the following questions: $1 \times 10 = 10$

- a) Explain the working principle of a color CRT monitor using Beam penetration method & Shadow-Mask method. $5 \times 2 = 10$
- b) Develop algorithm to clip a line using Cohen-Sutherland line clipping algorithm.
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B.Sc. 6th Semester (Honours) Examination, 2025

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Computer Graphics

(CBCS NEW 2022-23)

Time: 1 Hr 15 Min

Full Marks: 25

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Candidates are required to give their answer in their own words as far as practicable.*

Answer the questions unit wise as instructed

UNIT- I

1. Answer any five of the following questions: $5 \times 1 = 5$

- a) Name the scientist known as father of Computer Graphics
- b) What is frame buffer?
- c) Write two examples of scan converted objects.
- d) What is aliasing?
- e) What is the role of electron gun in a CRT monitor?
- f) Name a hidden surface removal algorithm.
- g) Name some issues in geometric modelling.
- h) Name a rigid body transformation that moves objects without deformation.

UNIT - II

2. Answer any two of the following questions: $2 \times 5 = 10$

- a) Write an algorithm to fill a polygon using flood fill.
- b) Consider the line from (0, 0) to (5, 6). Use Bresenham algorithm to rasterize this line.
- c) Write an algorithm to generate Bezier curve.

[Turn Over]

- d) Obtain the composite transformation matrix for general fixed point scaling.

UNIT - III

3. Answer any one of the following questions: $1 \times 10 = 10$

- a) Translate the triangle with co-ordinates A(2,5), B(7,10) and C(10,2) by 3 units in x-direction and 4 units in y-direction. Apply the shearing transformation to a square of vertices A(0, 0), B(1, 0), C(1, 1), D(0, 1), given that shear parameter value is 0.5 relative to the line $Y_{ref} = -1$. $5+5=10$
- b) Develop an algorithm to clip a line using Liang Barsky line clipping algorithm.

$$x, y \text{ incre}$$
$$m = \frac{dy}{dx} = \frac{y_p - y_q}{x_p - x_q}$$

Bankura Sammilani College

B.Sc. 6th Semester (Honours) Internal Assessment Examination 2025
Computer Science
Core-14 (Computer Graphics)

Answer any five questions:

2x5=10

1. What is pixel?
2. Differentiate between raster-scan and random-scan display.
3. What is shearing?
4. What do you mean by aspect ratio?
5. What is clipping?
6. Write two applications of computer graphics.
7. Write down drawbacks of DDA Line drawing algorithm.



B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61522

Course Code: SH/CSC/602/C-14

Computer Graphics (Practical)

(CBCS NEW 2022-23)

Time: 2 Hours

Full Marks: 15

*The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.*

(LNB + Viva = 05 Marks, Experiment = 10 Marks)

Perform any one experiment from the following:

- ✓ 1. Write a program to implement boundary fill algorithm.
 2. Write a program to implement DDA line drawing algorithm.
 3. Write a program to implement DDA circle drawing algorithm.
 4. Write a program to clip a line using Liang-Barsky line clipping algorithm.
 5. Write a program to clip a polygon using Sutherland Hodgeman algorithm.
 6. Write a program to apply 2D translation on a 2D object.
 7. Write a program to apply 3D rotation on a 3D object.
 8. Write a program to draw Bezier curve.
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Bankura Sammilani College

B.Sc. 6th Semester (Honours) Mock Test -2025
Computer Science
Computer Graphics (C-14)

Full Marks 25
Hours

Time: 1:15

Unit-I

1. Answer **any five** questions:

1x5=5

- ☒ a. What is clipping?
- ☒ b. What do you mean by frame buffer?
- ☒ c. What is translation?
- ☒ d. Differentiate between raster-scan and random-scan display.
- ☒ e. Name a hidden surface removal algorithm.
- ☒ f. What do you mean by vanishing point?
- ☒ g. What is parallel projection?

Unit-II

2. Answer **any two** questions:

5x2=10

- ☒ a. Write down the Bresenham's line drawing algorithm. Using this algorithm find out points of a line with end points (1, 4) and (4, 5).
- ☒ b. Define Bezier curve. Write some characteristics of this curve.
- c. What is 2D shearing? Obtain the composite transformation matrix for general fixed point scaling.
- d. Write and explain an algorithm used for hidden surface removal.

Unit-III

3. Answer **any one** question:

10x1=10

- a. Develop Cohen-Sutherland line clipping algorithm.
- b. Name three basic geometric transformations. Describe every such transformation for 3D objects.

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